Cognitive Load Modulates the Neural Mechanisms of Syntax–Plausibility Integration: An fMRI Study

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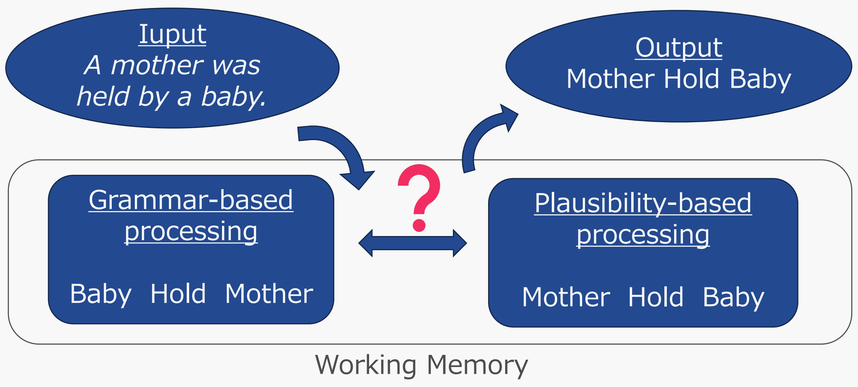
Take-home Message

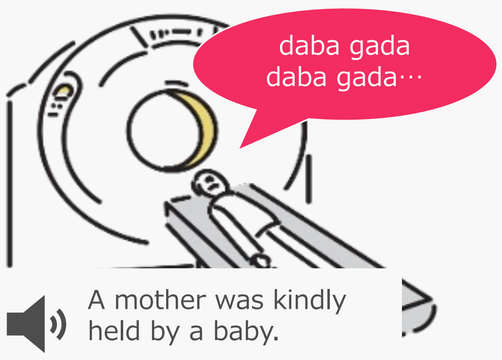
How do resource constraints modulate syntax–plausibility integration?

- ✓ ***Under limited cognitive resources, the LIFG pars triangularis (LIFGtri) adjusts the memory constraints on integration of syntactic and plausibility information.***
- ✓ ***Participants with higher WMC did NOT require increased activation in LIFGtri when processing implausible sentences during the dual-task condition.***

Research Question



- ○ **Is the integration of syntax–plausibility information modulated by resource constraints "during sentence processing"?**
- ○ **What neural mechanisms support sentence processing "under limited cognitive resources"?**

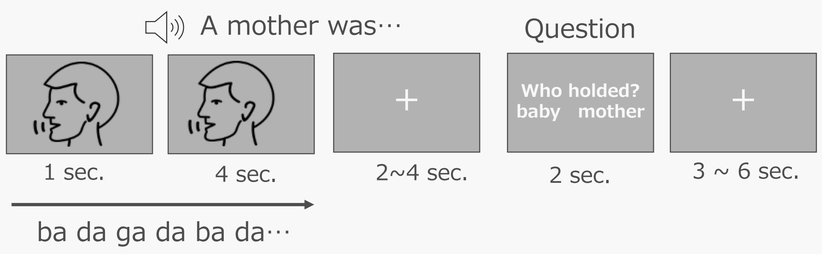


Methods

2 × 2 event-related design

Factor #1: Plausibility
- ○ **Plausible** — **A baby was held by a mother.**
- ○ **Implausible** — **A mother was held by a baby.**

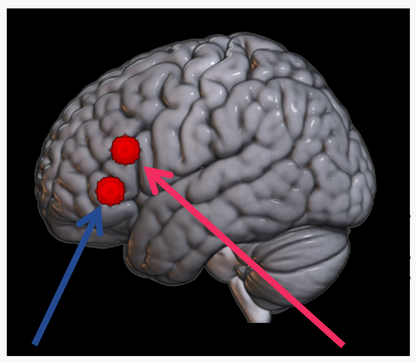
Factor #2: Secondary Task
- ○ **NoTask** — **only listening setences**
- ○ **Articulation** — **listening sentences during whispering "da ba ga da…"**



Participants
- ○ 62 native Japanese speakers (four participants were excluded)
- ○ Age: Mean = 21.81, SD = 3.26
- ○ 40 males, 22 females
- ○ Right-handed

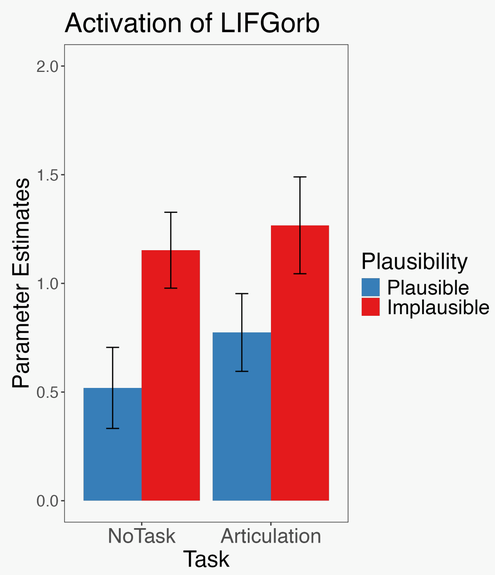
Stimuli
- ○ 48 target stimuli
- ○ 48 filler stimuli (plausible sentences with various constructions)
- ○ All Japanese

fROI Analysis

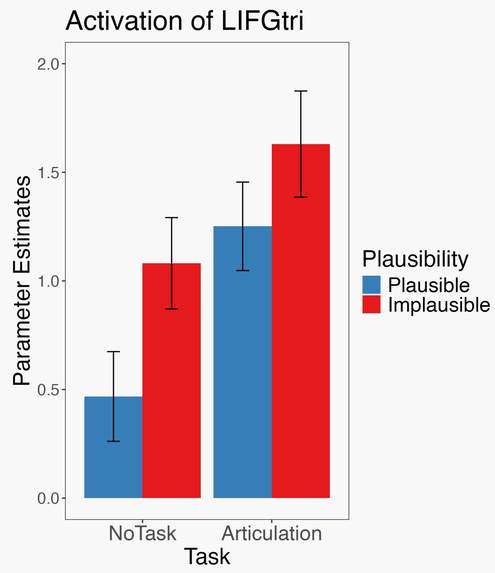


Coordinates from Sassa+ (2007)
8mm radium sphere
Activation ~ Plausibility * Task + (1|Participant)

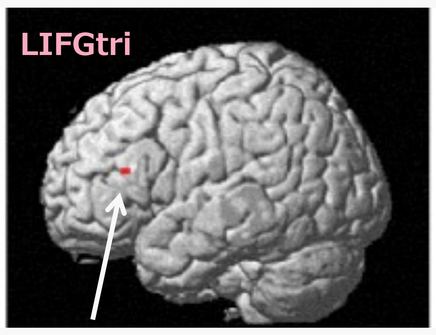
Implausible > Plausible **

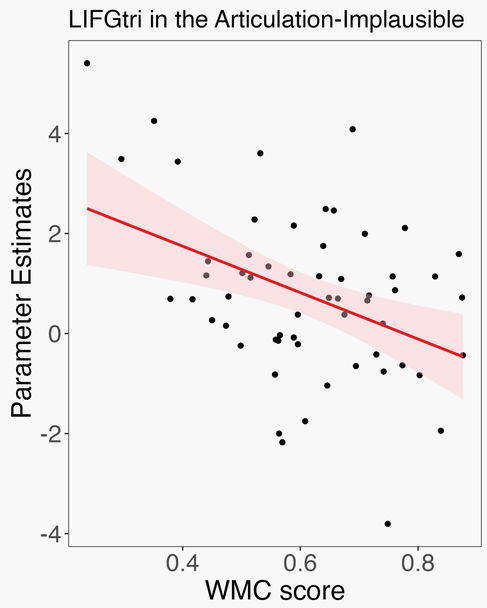


Implausible > Plausible **
Articulation > NoTask ***



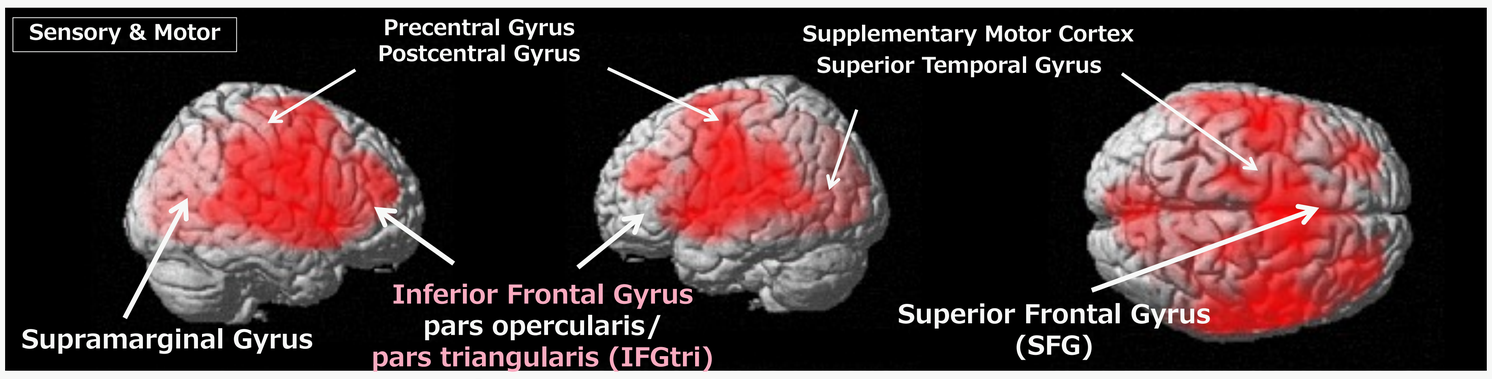
Working Memory Capacity





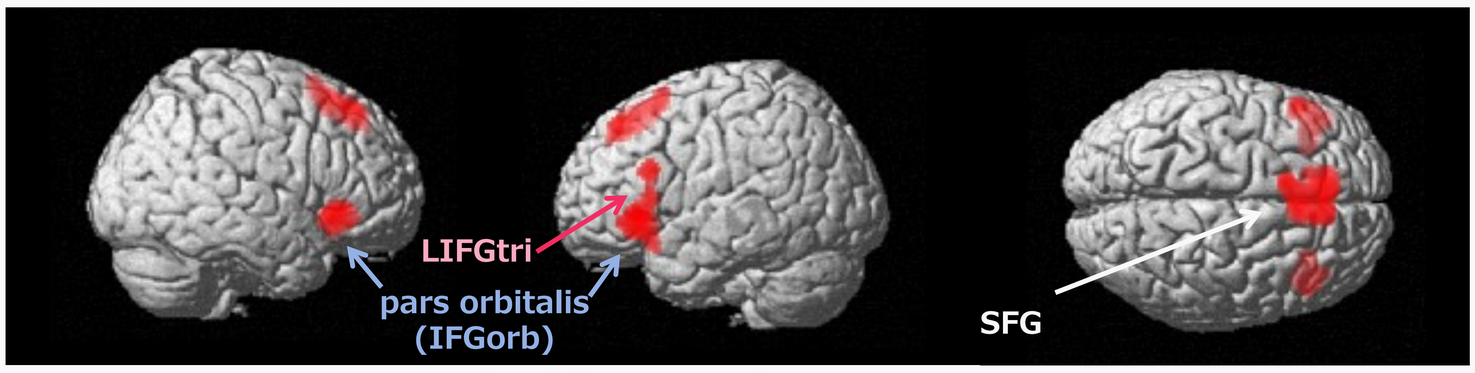
Whole Brain Analysis

Articulation > NoTask



→ **Verbal working memory** (Deschamps+ 2014)
→ **Sentence processing** (Fedorenko+ 2011; Bulut 2023)
→ **Verbal working memory** (Boisgueheneuc+ 2006; Nissim+ 2017)

Implausible > Plausible



→ **Syntax – Plausibility Integaration** (Sassa+ 2007; Ye & Zhou 2009)
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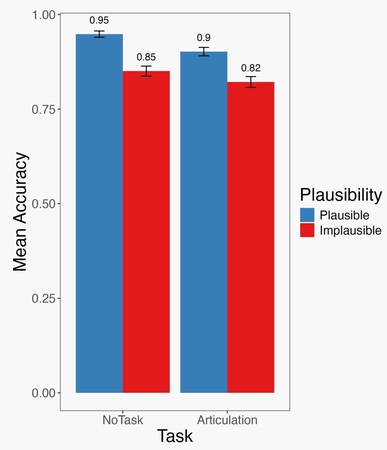
$p < 0.001$, cluster-size FWE $p < 0.05$

Accuracy



Plausible > Implausible ***
NoTask > Articulation **

- Correctness ~ Plausibility * Task + (1|Participant)
- Generalized linear mixed-effect model (family = binomial)

References

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